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Read the following rules carefully:

- Write your name and ID number in the solution you submit.
- Please sign and submit the code of honor in the exam with your solutions. The exam cannot be graded without a signed code of honor. You are supposed to do all of the problems on your own without receiving help from others.
- Cheating in the exam will not be tolerated. If you are caught cheating, you will be reported to the authorities. The recommendation of the instructor will be at least an F in the course in such cases.
- Do not post any questions on Piazza about the exam. The TAs have been instructed to refrain from answering questions about the midterm. In case of ambiguity or problems in the questions, just do your best.
- The use of generative AI is prohibited in answering the questions.
- Problems are not sorted in terms of difficulty. Please avoid guess work and long and irrelevant answers.
- Instructions on submitting the solutions to paper and pencil and coding problems will be provided shortly by the TAs.
- Show all your work and your final answer. Showing only the final answer of a question may not receive full credit and you must show your solution and reasoning behind the answer. Simplify your answer as much as you can.
- The exam has 9 questions, 17 pages, and a total of 100 points.
- The submission deadline for this midterm is 11:59 PM, Tuesday, July 1, 2025.
- As this is a take home exam that extends over several days, OSAS students DO NOT receive extra time, per OSAS guidelines.
- Any change in the midterm (paper and pencil or coding) after the deadline is considered late submission. One second late is late. The midterm is graded based on *when it was submitted, not when it was finished*. The midterm can be submitted up to three days late, with 10% penalty per late day. Homework late days cannot be used for the midterm.
- Submission after the grace period will receive a zero. One second late is late.

Grading Breakdown

Problem	Score	Earned
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
9	20	
Total	100	

Honor Code

I pledge on my honor that I have not given or received any unauthorized assistance on this examination.

Name: Abhishek Soundalgekar

Signature: 

1. Assume that in a regression problem with one independent variable $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, where the sample size n is 18 and the Pearson correlation r between X and Y is 0.36. Test at $\alpha = 0.01$ the null hypothesis $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$.

Note: Contrary to what you might think, you DO NOT need a t-table for this problem or any other information. Everything you need to solve this problem is given in this exam, please DO NOT double-check.

1.] Given: $R^2 = 0.36$ for simple linear model?

$$r^2 = 0.36$$

$$\alpha = 0.01$$

$$n = 18$$

no. of predictors $p = 1$

We have $F = \frac{(TSS - RSS)}{p} \div \frac{RSS}{(n - p - 1)}$ (substituting $p = 1$ we get)

$$F = \frac{TSS - RSS}{RSS / (n - 2)} \quad [\text{eq}^n 1]$$

We know $R^2 = r^2$ for simple linear model.

$$r^2 = \frac{TSS - RSS}{TSS} \quad [\text{eq}^n 2] \quad [RSS = TSS(1 - r^2)]$$

Substituting equation 2 in eqⁿ 1 we have:

$$F = \frac{r^2 \times TSS}{RSS / (n - 2)} = \frac{r^2 \times TSS}{TSS(1 - r^2) / (n - 2)} \quad (\text{from eq}^n 2)$$

$$F = \frac{r^2 (n - 2)}{(1 - r^2)}$$

Substituting value of $\alpha = 0.36$ we have

$$F = \frac{0.1296 \times 16}{0.8704}$$

$$F = 2.382$$

So for $df_1 = 1$, $df_2 = 16$, $\alpha = 0.01$

~~F~~

$$F_{1,16,0.01} = 8.53$$

Since $2.382 < 8.53$

we fail to reject the Null hypothesis

$H_0 : \beta_1$

Another Method for Question 1.

Given:

Regression Equation: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

sample size: $n = 18$

Pearson correlation between X and Y: $r = 0.36$

Test at significance level: $\alpha = 0.01$

Hypothesis: $H_0: \beta_1 = 0$ Vs. $H_1: \beta_1 \neq 0$

→ the appropriate test statistic

When testing the significance of the slope in simple linear regression, the test statistic can be expressed in terms of the correlation coefficient r :

$$t = r \sqrt{\frac{n-2}{1-r^2}}$$

→ We plug in the given values $n = 18, r = 0.36$

numerator: $n-2 = 18-2 = 16$

denominator: $1-r^2 = 1-(0.36)^2 = 1-0.1296 = 0.8704$

On substitution:

$$t = 0.36 \times \sqrt{\frac{16}{0.8704}}$$

$$t = 0.36 \times \sqrt{18.382} = 0.36 \times 4.29$$

$$t = 0.36 \times 4.29 \approx 1.54$$

We have two options when we have one predictor variable :

- (i) t -statistic with $n-2$ degrees of freedom
- (ii) F -statistic with $(1, n-2)$ degrees of freedom

Considering F -statistics with $(1, n-2)$

$$F_{1, (n-2)} = F_{1, 16} = 8.53 \quad (\text{from given table})$$

we know $F = t^2$

At $\alpha = 0.001$, the t -critical value is :

$$t_{\text{critical}} = \sqrt{F_{\text{critical}}} = \sqrt{8.53} = 2.9206$$

We can reject H_0 only if $|t| > t_{\text{critical}}$
But we got $t = 1.54$.

We can observe that $|t| < |t_{\text{critical}}|$

$\therefore$ We fail to reject H_0 [Null Hypothesis].

$\therefore$ At $\alpha = 0.001$ we can not reject the null hypothesis.

Therefore we fail to reject the null hypothesis

2. Consider the following unusual logistic regression model for binary classification:

$$\Pr(Y = 1|X_1, X_2) = \frac{\beta_0|X_1|^{\beta_1}|X_2|^{\beta_2}}{1 + \beta_0|X_1|^{\beta_1}|X_2|^{\beta_2}}$$

we fitted the model to a very large dataset using Maximum Likelihood and found the following coefficient estimates and standard errors:

	Coefficient	Standard Error
$\hat{\beta}_0$	1.1	0.2
$\hat{\beta}_1$	2.3	0.1
$\hat{\beta}_2$	0.93	0.7

- (a) Write down the log odds model for this unusual logistic regression model.
- (b) Find a 90% Confidence Interval for β_i 's.
- (c) Test at $\alpha = 0.03$ the statistical significance of $\hat{\beta}_i$'s.

2. a] Writing the log-odds model.

For any logistic regression model

$$\log\left(\frac{P}{1-P}\right) = \text{linear predictor}$$

According to our model:

$$P = \frac{\beta_0 | X_1 |^{\beta_1} | X_2 |^{\beta_2}}{1 + \beta_0 | X_1 |^{\beta_1} | X_2 |^{\beta_2}}$$

$$\frac{P}{1-P} = \beta_0 | X_1 |^{\beta_1} | X_2 |^{\beta_2}$$

$$\therefore \log\left(\frac{P}{1-P}\right) = \log(\beta_0) + \beta_1 \log |X_1| + \beta_2 \log |X_2|$$

Here we plug in the numbers for the fitted model:

$$\log\left(\frac{P}{1-P}\right) = \log(1.1) + 2.3 \log |X_1| + 0.93 \log |X_2|$$

2.b] The formula for a confidence interval is :

$$\hat{\beta}_i \pm z^* \cdot SE(\hat{\beta}_i)$$

For 90% CI, $z^* = 1.645$

$$\rightarrow \beta_0 : 1.1 \pm 1.645 \times 0.2 = 1.1 \pm 0.329 = [0.771, 1.429]$$

$$\rightarrow \beta_1 : 2.3 \pm 1.645 \times 0.1 = 2.3 \pm 0.1645 = [2.1355, 2.4645]$$

$$\rightarrow \beta_2 : 0.93 \pm 1.645 \times 0.7 = 0.93 \pm 1.1515 = [-0.2215, 2.0815]$$

2.c] Statistical significance at $\alpha = 0.03$ (two sided test)

Null hypothesis for each $H_0: \beta_i = 0$

The test statistic is:

$$Z = \frac{\hat{\beta}_i}{SE(\hat{\beta}_i)}$$

For a two-sided test at $\alpha = 0.03$, cutoff is approximately $Z_{0.015} = 2.17$.

For each coefficient:

$$\beta_0: Z = \frac{1.1}{0.2} = 5.5$$

$|5.5| > 2.17 \Rightarrow$ statistically significant

$$\beta_1: Z = \frac{2.3}{0.1} = 23.0$$

$|23.0| > 2.17 \Rightarrow$ statistically significant

$$\beta_2: Z = \frac{0.93}{0.7} \approx 1.33$$

$|1.33| < 2.17 \Rightarrow$ statistically NOT significant

Therefore:

both β_0 & β_1 are statistically significant at the 3% level. β_2 is not statistically significant

3. Consider the regression problem with an unusual regularizer:

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \beta_3 x_{i3})^2 + \lambda \{(\beta_1 - \beta_2)^2 + (\beta_3 - \beta_2)^2\}$$

- (a) Write down the problem as a constrained minimization problem.
- (b) Explain what happens to β_i 's when λ is very large.

3.a] Constrained Minimization problem.

- The regularizer $(\beta_1 - \beta_2)^2 + (\beta_3 - \beta_2)^2$ penalizes the model if the coefficients for the three features differ too much.
- In optimization, adding a regularization term is equivalent to putting a constraint on the quality being penalized.

We can rewrite it as:

$$\text{Minimize } \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \beta_3 x_{i3})^2$$

Subject to

$$(\beta_1 - \beta_2)^2 + (\beta_3 - \beta_2)^2 \leq t$$

where t is a non-negative constant that determines how strictly we enforce the similarity between the coefficients (t gets smaller as λ increases).

$\therefore$ Minimize the sum of squared errors subject to a bound on how different the coefficients can be, specifically, the sum of the squared differences from β_2 must not exceed t .

- 3 b] → The parameter λ controls how much importance we give to the penalty term.
- As λ increases, the cost for the coefficients being different from each other increases.
 - When λ is extremely large, the optimization is forced to make the penalty as small as possible, otherwise the total loss would become huge.

* The penalty : $(\beta_1 - \beta_2)^2 + (\beta_3 - \beta_2)^2$

→ This is minimized when both the terms are zero, i.e., when $\beta_1 = \beta_2$ and $\beta_3 = \beta_2$.

As $\lambda \rightarrow \infty$:

- Any difference between the coefficients is infinitely penalized.
- The solution must be : $\beta_1 = \beta_2 = \beta_3$
- The model essentially treats all three predictors as having the same impact on the output.

∴ When λ is very large, the regularization dominates the optimization. This forces the 3 coefficients to be equal, regardless of what the data says. The regression problem effectively reduces to a model where all 3 variables are treated as having an identical effect, so the fitted equation becomes : $y_i \approx \beta_0 + \beta^* (x_{i1} + x_{i2} + x_{i3})$

Here β^* is the shared coefficient value.

4. A tech company wants to classify customer reviews as either *Positive* or *Negative*. You are given the following dataset extracted from past reviews:

Review ID	Word: “Good”	Word: “Cheap”	Word: “Fast”	Label
1	1	1	0	Positive
2	0	1	1	Negative
3	1	0	1	Positive
4	0	1	0	Negative
5	1	1	1	Positive

- (a) A new review contains the following features:

$$\text{Good} = 1, \quad \text{Cheap} = 1, \quad \text{Fast} = 0.$$

Using the Naïve Bayes classifier, predict whether this review is *Positive* or *Negative*. Use Laplace Smoothing, except in calculating the prior probabilities. Show all calculations step-by-step.

- (b) What assumptions does the Naïve Bayes classifier make about the features, and why might these assumptions be unrealistic in real-world scenarios?

4.a] Naïve Bayes Prediction (with Laplace Smoothing)

→ Computed Prior Probabilities (no smoothing for priors)

Total reviews : 5

Positive reviews : 3 (IDs 1, 3, 5)

Negative reviews : 2 (IDs 2, 4)

$$P(\text{Positive}) = \frac{3}{5} = 0.6$$

$$P(\text{Negative}) = \frac{2}{5} = 0.4$$

→ Computed Likelihoods with Laplace Smoothing

Laplace Smoothing : Add 1 to numerator, add number of possible feature values (2) to denominator.

- For each class, count number of instances.
- For each feature value, count occurrences within class, then apply smoothing.

Positive Class ($n=3$)

Reviews : 1, 3, 5

• Good = 1

Appears in reviews 1, 3, 5 $\Rightarrow$ count = 3

$$P(\text{Good} = 1 | \text{Pos}) = \frac{3+1}{3+2} = \frac{4}{5}$$

- Cheap = 1:

Appears in reviews 1, 5 $\Rightarrow$ count = 2

$$P(\text{Cheap} = 1 | \text{Pos}) = \frac{2+1}{3+2} = \frac{3}{5}$$

- Fast = 0:

Appears in review 1 $\Rightarrow$ count = 1

$$P(\text{Fast} = 0 | \text{Pos}) = \frac{1+1}{3+2} = \frac{2}{5}$$

Negative class ($n=2$)

Reviews: 2, 4

- Crood = 1:

Appears in none $\Rightarrow$ count = 0

$$P(\text{Crood} = 1 | \text{Neg}) = \frac{0+1}{2+2} = \frac{1}{4}$$

- Cheap = 1:

Appears in both $\Rightarrow$ count = 2

$$P(\text{Cheap} = 1 | \text{Neg}) = \frac{2+1}{2+2} = \frac{3}{4}$$

- Fast = 0:

Appears in review 4 $\Rightarrow$ count = 1

$$P(\text{Fast} = 0 | \text{Neg}) = \frac{1+1}{2+2} = \frac{2}{4} = \frac{1}{2}$$

→ Posterior (Unnormalized) Probabilities calculation:

For Positive:

$$P(\text{Positive}) \times P(\text{Good}=1 | \text{Pos}) \times P(\text{Cheap}=1 | \text{Pos}) \times P(\text{Fast}=0 | \text{Pos})$$

$$= 0.6 \times \frac{4}{5} \times \frac{3}{5} \times \frac{2}{5}$$

$$= 0.1152$$

For Negative:

$$P(\text{Negative}) \times P(\text{Good}=1 | \text{Neg}) \times P(\text{Cheap}=1 | \text{Neg}) \times P(\text{Fast}=0 | \text{Neg})$$

$$= 0.4 \times \frac{1}{4} \times \frac{3}{4} \times \frac{1}{2}$$

$$= 0.0375$$

As we can observe $0.1152 > 0.0375$

i.e., Positive > Negative.

∴ The Naive Bayes classifier predicts 'Positive'.

4.6] Naïve Bayes Assumptions & Limitations

Assumptions :

- i) The Naïve Bayes model assumes all features (words : 'Fast', 'cheap', 'good') are conditionally independent given the class label.
- ii) In other words, knowing the value of one word does not give you any information about the value of another, if you already know the label.

* This is often unrealistic :

- i) In real language data, words in reviews are often correlated. For example 'cheap' and 'good' might often appear together in positive reviews.
- ii) This independence assumption may ignore useful relationships and patterns between words (features), leading to poorer predictions in practice.
- iii) Despite the simplicity, Naïve Bayes often works & reasonably well, but its strong independence assumption is rarely true for text data.

5. Consider the following data set for regression:

Index	X	Y
1	0	1
2	-2	2
3	-1	-2

- (a) Show all possible bootstrap samples of the dataset that have the same size as this dataset. Note that permutations of the same data set are considered the same dataset, for example $\{1, 2, 3\}$ and $\{2, 3, 1\}$ are the same dataset.
- (b) Construct a KNN regression model for all bootstrap samples in part 5a that contain the data point $(x, y) = (0, 1)$ with $K = 2$ and predict the label for the test point $x^* = 0$ by averaging the predictions of those bootstrap models.

5 a] All possible bootstrap samples (with duplicates, Order ignored)

A bootstrap sample is obtained by sampling n points with replacement from a dataset of size n , where the sample may include repeated data points.

Here the data size $n=3$. Each sample consists of 3 points (possibly with repeats), chosen from points 1, 2, 3.

* Let's enumerate all possible multisets (since order does not matter).

1. All from one point:

$\{1, 1, 1\}$ $\{2, 2, 2\}$ $\{3, 3, 3\}$

2. Two of one, one of another:

$\{1, 1, 2\}$, $\{1, 1, 3\}$, $\{2, 2, 1\}$, $\{2, 2, 3\}$, $\{3, 3, 1\}$, $\{3, 3, 2\}$

As $\{1, 1, 2\}$ & $\{2, 1, 1\}$ are same so I kept just one of each.

3. All distinct:

$\{1, 2, 3\}$

So, the complete list of unique bootstrap samples is:

$\{1, 1, 1\}, \{2, 2, 2\}, \{3, 3, 3\}, \{1, 1, 2\}, \{1, 1, 3\}, \{2, 2, 1\},$
 $\{2, 2, 3\}, \{3, 3, 1\}, \{3, 3, 2\}, \{1, 2, 3\}, \{1, 2, 2\},$
 $\{1, 3, 3\}, \{2, 3, 3\}, \{2, 1, 1\}, \{3, 1, 1\}, \{3, 2, 2\}$
 $\{1, 3, 2\}$

There is an issue with this approach, actually the possible unique multisets of size 3 from 3 items are:

Final list:

$\{1, 1, 1\}, \{2, 2, 2\}, \{3, 3, 3\}, \{1, 1, 2\}, \{1, 1, 3\},$
 $\{2, 2, 3\}, \{1, 2, 2\}, \{1, 3, 3\}, \{2, 3, 3\}, \{1, 2, 3\}.$

The issue was with

- $\{2, 2, 1\}$ (already counted as $\{1, 1, 2\}$) so ignore.
- $\{3, 3, 1\}$ (already counted as $\{1, 1, 3\}$) so ignore.
- $\{3, 3, 2\}$ (already counted as $\{2, 2, 3\}$) so ignore.

5 b] KNN Regression with $K=2$, only for samples containing $(0,1)$ (i.e., point 1)

→ Bootstrap samples that include $(0,1)$ are:
(from part 5a, the ones containing at least one)
 $\{1,1,1\}$, $\{1,1,2\}$, $\{1,1,3\}$, $\{1,2,2\}$, $\{1,3,3\}$, $\{1,2,3\}$.

There were more but these are ones containing at least one $(0,1)$; the rest are with only 2s and 3s.

→ For each sample, I will build KNN regressor and predict at $x^* = 0$ with $K=2$.

Sample 1: $\{1,1,1\}$

All points are $(0,1)$.

- The only observed value for x is 0, and all targets are 1.
- For KNN with $K=2$, prediction at $x^* = 0$ is the average of two nearest neighbours' y values. Both are 1.

Prediction: 1

Sample 2: $\{1,1,2\}$

Points: $(0,1)$ $(0,1)$ $(-2,2)$

Distance to $x^* = 0$

$(0,1) : 0$

$(0,1) : 0$

$(-2,2) : 2$

the two closest: $(0,1) (0,1)$

$$\text{Average } y = \frac{1+1}{2} = 1$$

Prediction: 1

Sample 3: $\{1,1,3\}$

Points: $(0,1) (0,1) (-1,-2)$

Distances: 0, 0, 1

Two nearest: $(0,1), (0,1)$

Average: 1

Prediction: 1

Sample 4: $\{1,2,2\}$

Points: $(0,1), (-2,2), (-2,2)$

Distances: 0, 2, 2

Two closest: $(0,1), (-2,2)$

Average $y: (1+2)/2 = 1.5$

Prediction = 1.5

Sample 5: $\{1,3,3\}$

Points: $(0,1), (-1,-2), (-1,-2)$

Distances: 0, 1, 1

Two nearest: $(0,1), (-1,-2)$

Average $y: (1 + (-2))/2 = -0.5$

Prediction = -0.5

Sample 6 = $\{1, 2, 3\}$

Points: $(0, 1), (-2, 2), (-1, -2)$

distances $(0, 2, -1)$

two nearest $(0, 1), (-1, -2)$

Average $y = (1 + (-2))/2 = -0.5$

Prediction: -0.5

→ Average predictions from these models:

Predictions: $1, 1, 1, 1.5, -0.5, -0.5$

$$\begin{aligned}\text{Mean Prediction} &= \frac{1 + 1 + 1 + 1.5 + (-0.5) + (-0.5)}{6} \\ &= \frac{3.5}{6} \approx 0.583\end{aligned}$$

Averaging these gives a final prediction 0.583

6. Assume that in a classification problem with one feature, the conditional distribution of the feature in class $k \in \{1, 2, 3\}$ is $f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp(-\frac{(x-\mu_k)^2}{2\sigma_k^2})$, where $\mu_k = \sigma_k = k, k \in \{1, 2, 3\}$. Also, assume that the prior probabilities of each class are $\pi_k = \frac{1}{3}, k \in \{1, 2, 3\}$. Build a multinomial regression model that yields the same decision boundaries as the Bayesian Quadratic Discriminant Analysis method for this three-class problem; that is, show $\Pr(Y = k|X = x)$ for all $k \in \{1, 2, 3\}$. Note that we want each class to have a vector of parameters associated with it, i.e. three parameter vectors. Hence, the posterior probabilities will be in the form:

$$\Pr(Y = k|X = x) = \frac{e^{g_k(x)}}{e^{g_1(x)} + e^{g_2(x)} + e^{g_3(x)}}$$

6

Given

- 3 classes $K \in \{1, 2, 3\}$
- class k :

$$f_k(x) = \frac{1}{\sqrt{2\pi} \sigma_k} \exp\left(\frac{-(x - \mu_k)^2}{2\sigma_k^2}\right)$$

here $\mu_k = k$, $\sigma_k = k$

Priors : $\pi_k = \frac{1}{3}$

We have to construct a multinomial regression model (multiclass logistic regression or softmax regression) with decision boundaries matching the Bayes classifier, i.e., QDA for these Gaussian distributions.

→ Posterior Probability via Bayes Theorem

$$P(Y=K | X=x) = \frac{\pi_k f_k(x)}{\sum_{j=1}^3 \pi_j f_j(x)}$$

→ Expressing $f_k(x)$ in Exponential form.

$$f_k(x) = \frac{1}{\sqrt{2\pi} k} \exp\left(\frac{-(x-k)^2}{2k^2}\right)$$

Expanding the exponent we get :

$$-\frac{(x-k)^2}{2k^2} = -\frac{(x^2 - 2kx + k^2)}{2k^2} = \frac{-x^2}{2k^2} + \frac{kx}{k^2} - \frac{k^2}{2k^2}$$

$$= \frac{-x^2}{2k^2} + \frac{x}{k} - \frac{1}{2}$$

$$f_k(x) = \frac{1}{\sqrt{2\pi} k} \exp\left(\frac{-x^2}{2k^2} + \frac{x}{k} - \frac{1}{2}\right)$$

→ Posterior as softmax form:

Multinomial regression model uses:

$$P(Y=k | X=x) = \frac{e^{g_k(x)}}{\sum_{j=1}^3 e^{g_j(x)}}$$

here $g_k(x)$ is the discriminant function for class k .

From Bayes & QDA, the discriminant is the log of the numerator (ignoring constants common to all classes):

$$g_k(x) = \log\left(\frac{1}{\sqrt{2\pi} k}\right) - \frac{x^2}{2k^2} + \frac{x}{k} - \frac{1}{2}$$

$$= \frac{-1}{2} \log(2\pi) - \log k - \frac{x^2}{2k^2} + \frac{x}{k} - \frac{1}{2}$$

Since the $\frac{-1}{2} \log(2\pi)$ and $-\frac{1}{2}$ are the same

for all k , going to drop these won't matter, so I drop them, they cancel out in the softmax denominator and numerator.

so use the 'reduced' discriminant functions:

$$g_k(x) = -\log k - \frac{x^2}{2k^2} + \frac{x}{k}$$

→ softmax form:

Multinomial Regression Model is:

$$P(Y=k | X=x) = \frac{\exp\left(-\log k - \frac{x^2}{2k^2} + \frac{x}{k}\right)}{\sum_{j=1}^3 \exp\left(-\log j - \frac{x^2}{2j^2} + \frac{x}{j}\right)}$$

for $k \in \{1, 2, 3\}$.

Each class has its own discriminant function:

$$g_k(x) = -\log k - \frac{x^2}{2k^2} + \frac{x}{k}$$

this multinomial logistic regression will have the same decision boundaries as the QDA Bayes classifier for these class-conditional Gaussians.

7. Consider the following dataset:

Index	X	Y
1	0	1
2	1	-1
3	-1	0

Find the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for a simple linear regression model $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. Then determine the mean-squared test error of your model on the following test set:

Index	X	Y
1	2	0
2	-1	-1

7.] Least squares Estimates $\hat{\beta}_0$ and $\hat{\beta}_1$

Formula for simple linear regression:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

First calculating the means $\bar{x}$ & $\bar{y}$.

$$\bar{x} = \frac{0 + 1 + (-1)}{3} = 0$$

$$\bar{y} = \frac{1 + (-1) + 0}{3} = 0$$

Numerator & denominator for $\hat{\beta}_1$:

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i$$

$$\text{For } i=1 : x_1 y_1 = 0 \times 1 = 0$$

$$\text{For } i=2 : x_2 y_2 = 1 \times -1 = -1$$

$$\text{For } i=3 : x_3 y_3 = -1 \times 0 = 0$$

$$\text{Sum} = -1$$

$$\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2$$

For $i=1$: $0^2 = 0$

For $i=2$: $1^2 = 1$

For $i=3$: $(-1)^2 = 1$

Sum = 2

$$\therefore \hat{\beta}_1 = \frac{-1}{2} = -0.5$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0 - (-0.5) \times 0 = 0$$

→ Fitted Regression Model

$$\hat{y}_i = 0 - 0.5x_i = -0.5x_i$$

→ Test Set

Index	x_i^*	y_i^*
1	2	0
2	-1	-1

→ Predictions on the test set

• For $x_1^* = 2$: $\hat{y}_1^* = -0.5 \times 2 = -1$

• For $x_2^* = -1$: $\hat{y}_2^* = -0.5 \times (-1) = 0.5$

→ Mean squared test error:

$$MSE = \frac{1}{2} \left[(y_1^* - \hat{y}_1^*)^2 + (y_2^* - \hat{y}_2^*)^2 \right]$$

- For test point 1: $(0 - (-1))^2 = 1^2 = 1$
- For test point 2: $(-1 - 0.5)^2 = (-1.5)^2 = 2.25$

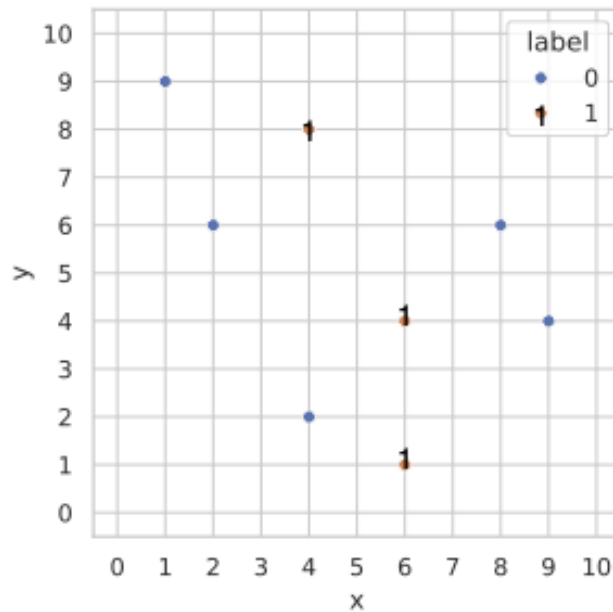
$$MSE = \frac{1}{2} (1 + 2.25) = \frac{3.25}{2} = 1.625$$

i. Final values:

$$\hat{\beta}_0 = 0, \quad \hat{\beta}_1 = -0.5$$

$$MS \text{ Test Error} = 1.625$$

8. Consider the following dataset with two real-valued features, x and y , and one binary output. [By Yue Wu]



We want to use 1-nearest-neighbor learning to predict the output using the input (x, y) , with a distance metric yet to be determined. Calculate the leave-one-out cross validation (LOOCV) error of 1-NN for each of the following distance metrics, and decide which distance metric is better. **Break ties in favor of class 0.**

- $d_1((x_1, y_1), (x_2, y_2)) = |x_1 - x_2|$,
- $d_2((x_1, y_1), (x_2, y_2)) = |y_1 - y_2|$,
- $d_3((x_1, y_1), (x_2, y_2)) = \max(|x_1 - x_2|, |y_1 - y_2|)$.
- $d_4((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

8.]

Four distance Metrics

Given two points $(x_1, y_1), (x_2, y_2)$:

$$d_1 = |x_1 - x_2| \text{ (X-distance)}$$

$$d_2 = |y_1 - y_2| \text{ (Y-distance)}$$

$$d_3 = \max(|x_1 - x_2|, |y_1 - y_2|) \text{ (Chebyshev)}$$

$$d_4 = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \text{ (Euclidean)}$$

→ LOOCV for Each Metric

For each data point :

Remove it, find its nearest neighbour among the remaining 7 points using the given metric then assign its label to the test point and record if the prediction matches the true label and last if there is a tie, break in favour of class 0.

→ LOOCV for d_1 (X-distance only)

Point 1 : $(1, 9), 0$

Nearest x : $(2, 6)$ (distance 1, label 0)

Prediction : 0 (correct)

Point 2 : $(2, 6), 0$

Nearest x : $(1, 9)$ or $(4, 2)$ (distance 1 or 2, both label 0)

Prediction : 0 (correct)

Point 3 : $(4, 2), 0$

Nearest n : $(6, 1)$ or $(4, 8)$ or $(4, 2)$ (itself)

but itself is left out.

Next closest : $(6, 1)$ or $(4, 8)$ distance 2 or 0, both label 1 or 0

For this :

$(2, 6) : n = 2$ (distance 2, label 0)

$(6, 1) : n = 6$ (dist. 2, label 0)

$(4, 8) : n = 4$ (dist. 0, label 1)

closest : $(4, 8)$, label 1

Prediction : 1 (incorrect)

Point 4 : $(8, 6), 0$

Nearest n : $(9, 4)$ (distance 1, label 0)

Prediction : 0 (correct)

Point 5 : $(9, 4), 0$

Nearest n : $(8, 6)$ (distance 1, label 0)

Prediction : 0 (correct)

Point 6 : $(4, 8), 1$

Nearest n : $(4, 2)$ (distance 0, label 0)

Prediction : 0 (incorrect)

Point 7: $(6, 1), 1$

Nearest n: $(4, 2)$ (distance 2, label 0) or $(6, 4)$ (dist 0, lab 1)

closest: $(6, 4)$, label 1

Prediction: 1 (correct)

Point 8: $(6, 4), 1$

Nearest n: $(6, 1)$ (dist 0, lab. 1)

Pred: 1 (correct)

Errors: Point 3 & 6 (2 errors)

→ LOOCV for d_2 (V-distance only)

Point 1: $(1, 9), 0$

N. y: $(4, 8)$ (d 1, l 1)

Pred: 1 (incorrect)

Point 2: $(2, 6), 0$

N. y: $(8, 6)$ (d. 0, label 0)

Pred: 0 (correct)

Point 3: $(4, 2), 0$

Nearest y: $(6, 1)$ (d 1, l 1)

Pred: 1 (incorrect)

Point 4: $(8, 6), 0$

Nearest y : $(2, 6)$ (dist 0, label 0)

Pred: 0 (correct)

Point 5: $(9, 4), 0$

Nearest y : $(6, 4)$ (dist 0, label 0)

Pred: 1 (incorrect)

Point 6: $(4, 8), 1$

Nearest y : $(1, 9)$ (dist 1, label 0)

Pred: 0 (incorrect)

Point 7: $(6, 1), 1$

Nearest y : $(4, 2)$ (distance 1, label 0)

Pred: 0 (incorrect)

Point 8: $(6, 4), 1$

Nearest y : $(9, 4)$ (distance 0, label 0)

Pred: 0 (incorrect)

Errors: Points 1, 3, 5, 6, 7, 8 (6 errors)

→ LOOCV of d_3 (Chebyshev/Max)

Point 1: $(1, 9), 0$

Closest: $(4, 8)$ ($dx=3, dy=1, \max=3$)

Also $(2, 6)$: $dx=1, dy=3, \max=3$

Both distances = 3, $(2, 6)$: label 0

$(4, 8)$: label 1

tie-break at 0

Prediction: 0 (correct)

Point 2: $(2, 6), 0$

$(4, 8)$: $dx=2, dy=2, \max=2$

$(8, 6)$: $dx=6, dy=0, \max=6$

$(1, 9)$: $dx=1, dy=3, \max=3$

Closest: $(4, 8)$, label 1

Prediction: 1 (incorrect)

Point 3: $(4, 2), 0$

$(6, 1)$: $dx=2, dy=1, \max=2$ (label 1)

$(2, 6)$: $dx=2, dy=4, \max=4$

$(6, 4)$: $dx=2, dy=2, \max=2$ (label 1)

Closest: $(6, 1)$ or $(6, 4)$ both distance 2, both label 1;

tie to 0 (but no class 0 neighbor at same distance), so 1

Prediction: 1 (incorrect)

Point 4 : $(8, 6), 0$

$(9, 4)$: $dx=1, dy=2, \max=2$ (label 0)

$(2, 6)$: $dx=6, dy=0, \max=6$

$(6, 4)$: $dx=2, dy=2, \max=2$ (label 1)

closest : $(9, 4)$ or $(6, 4)$: both distance 2, label 0, and 1 ; tie-break to 0

Prediction : 0 (correct)

Point 5 : $(9, 4), 0$

$(8, 6)$: $dx=1, dy=2, \max=2$ (label 0) ←

$(6, 4)$: $dx=3, dy=0, \max=3$ (label 1)

closest : $(8, 6)$, label 0

Prediction : 0 (correct)

Point 6 : $(4, 8), 1$

$(1, 9)$: $dx=3, dy=1, \max=3$ (label 0)

$(2, 6)$: $dx=2, dy=2, \max=2$ (label 0)

$(8, 6)$: $dx=4, dy=2, \max=4$ (label 0)

closest : $(2, 6)$, label 0

Prediction : 0 (incorrect)

Point 7 : $(6, 1), 1$

$(4, 2)$: $dx=2, dy=1, \max=2$ (label 0)

$(6, 4)$: $dx=0, dy=3, \max=3$ (label 1)

closest : $(4, 2)$, label 0

Prediction : 0 (incorrect)

Point 8: $(6, 4), 1$

$(9, 4)$: $dx=3, dy=0, \max=3$ (label 0)

$(6, 1)$: $dx=0, dy=3, \max=3$ (label 1)

closest: $(9, 4)$ or $(6, 1)$ both 3, labels 0 and 1;

tie to 0

Prediction: 0 (incorrect)

Errors: points 2, 3, 6, 7, 8 (5 errors)

→ LOOCV for d_1 (Euclidean distance)

Point 1: $(1, 9), 0$

$(2, 6)$: $\sqrt{1^2 + 3^2} = \sqrt{10} = 3.16$, label 0

$(4, 8)$: $\sqrt{3^2 + 1^2} = \sqrt{10} = 3.16$, label 1

Both tie, choose class 0

Prediction: 0 (correct)

Point 2: $(2, 6), 0$

$(1, 9)$: $\sqrt{1^2 + 3^2} = 3.16$, label 0

$(4, 8)$: $\sqrt{2^2 + 2^2} = 2.83$, label 1

$(4, 2)$: $\sqrt{2^2 + 4^2} = 4.47$, label 0

Closest: $(4, 8)$, label 1

Prediction: 1 (incorrect)

Point 3 : $(4, 2), 0$

$$(6, 1) : \sqrt{2^2 + 1^2} = 2.24, \text{ label } 1$$

$$(2, 6) : \sqrt{2^2 + 4^2} = 4.47, \text{ label } 0$$

$$(6, 4) : \sqrt{2^2 + 2^2} = 2.83, \text{ label } 1$$

Closest : $(6, 1)$, label 1

Prediction : 1 (incorrect)

Point 4 : $(8, 6), 0$

$$(9, 4) : \sqrt{1^2 + 2^2} = 2.24, \text{ label } 0$$

$$(6, 4) : \sqrt{2^2 + 2^2} = 2.83, \text{ label } 1$$

Closest : $(9, 4)$, label 0

Prediction : 0 (correct)

Point 5 : $(9, 4), 0$

$$(8, 6) : \sqrt{1^2 + 2^2} = 2.24, \text{ label } 0$$

$$(6, 4) : \sqrt{3^2 + 0^2} = 3, \text{ label } 1$$

Closest : $(8, 6)$, label 0

Prediction : 0 (correct)

Point 6 : $(4, 8), 1$

$$(2, 6) : \sqrt{2^2 + 2^2} = 2.83, \text{ label } 0$$

$$(1, 9) : \sqrt{3^2 + 1^2} = 3.16, \text{ label } 0$$

$$(8, 6) : \sqrt{4^2 + 2^2} = 4.47, \text{ label } 0$$

Closest : $(2, 6)$, label 0

Prediction : 0 (incorrect)

Point 7 : $(6, 1), 1$

$$(4, 2) : \sqrt{2^2 + 1^2} = 2.24, \text{ label } 0$$

$$(6, 4) : \sqrt{0^2 + 3^2} = 3, \text{ label } 1$$

Closest : $(4, 2)$, label 0

Prediction : 0 (incorrect)

Point 8 : $(6, 4), 1$

$$(9, 4) : \sqrt{3^2 + 0^2} = 3, \text{ label } 0$$

$$(6, 1) : \sqrt{0^2 + 3^2} = 3, \text{ label } 1$$

Both : tie, break to class 0

Prediction : 0 (incorrect)

Errors : points 2, 3, 6, 7, 8 (5 errors)

Metric	LOOCV errors
d_1	2
d_2	6
d_3	5
d_4	5

lowest LOOCV error : d_1 (1-distance), with only 2 errors.

$\therefore$ the best distance metric for this dataset is

$d_1((x_1, y_1), (x_2, y_2)) = |x_1 - x_2|$ (distance in the x direction only), as it has the lowest leave-one-out error.

If there are ties, class 0 is chosen by default.

9. Programming Question: Predicting Housing Prices with Linear Regression

Dataset: `USA_Housing.csv`

link: <https://www.kaggle.com/datasets/vedavyasv/usa-housing>

In this problem, you will build a linear regression model to predict housing prices in the USA based on various features. You will use Python and `scikit-learn` for this task.

(a) Data Exploration and Pre-processing

- i. Load the `USA_Housing.csv` dataset into a pandas DataFrame.
- ii. Explore the dataset: First, display the first 5 rows of the DataFrame. Then, get a concise summary of the DataFrame, including data types and non-null values and generate descriptive statistics for the numerical columns. iv. Check for any missing values in the dataset and report their counts per column.
- iii. Identify the features (independent variables) and the target variable (dependent variable). The target variable is `Price`. All other numerical columns should be considered features.
- iv. **Feature Engineering:** Create a new feature called `Avg_Area_Room_Per_Person` by dividing `Avg. Area Number of Rooms` by `Avg. Area Number of Bedrooms`. Discuss the potential rationale for creating this feature. Consider the `Address` column. Although it's categorical, can you extract any numerical information from it that might be useful? (e.g., zip codes, if present, or length of the address string). If you extract a numerical feature, justify its potential relevance.
- v. **Outlier Detection:** Use a suitable method (e.g., Z-score or IQR) to identify potential outliers in the `Price` column. Visualize the distribution of the `Price` column before and after outlier handling (e.g., using box plots or histograms). Decide whether to remove or transform these outliers and justify your choice.

(b) Linear Regression Model

- i. Split your data into training and testing sets. Use 80% of the data for training and 20% for testing. Set `random_state` to 42 for reproducibility.
- ii. Create a Linear Regression model.
- iii. Train the model using the training data.
- iv. Make predictions on the test set.

(c) Regularization

- i. Implement and train two additional models: a Ridge Regression model and a Lasso Regression model.
- ii. For both Ridge and Lasso, experiment with at least three different values for the regularization parameter (α). Briefly explain the effect of α on the model.
- iii. Compare the coefficients learned by the standard Linear Regression, Ridge, and Lasso models. Discuss any noticeable differences, especially in terms of feature selection (Lasso).

(d) Model Evaluation

- i. Evaluate the performance of your linear regression model on the test set.
- ii. Calculate and report the following metrics for *all* trained models (Linear Regression, Ridge, Lasso):
 - A. Mean Absolute Error (MAE)
 - B. Mean Squared Error (MSE)
 - C. Root Mean Squared Error (RMSE)
 - D. R-squared (R^2) score
- iii. Interpret the meaning of each of these metrics in the context of predicting housing prices.
- iv. **Cross-Validation:** Perform k-fold cross-validation (e.g., with $k = 5$ or $k = 10$) on your best performing model (determined from section 3b). Then, Report the average and standard deviation of the R^2 score (or other chosen metric) across all folds. Discuss the advantages of using cross-validation over a single train-test split for evaluating model performance.

(e) Coefficients and Intercept

- i. Display the coefficients of the features and the intercept of your *standard* linear regression model.
- ii. Discuss what these coefficients indicate about the relationship between each feature and the housing price. For example, how does a one-unit increase in a particular feature affect the predicted housing price, holding other features constant?

Expected Output: Your submission should include:

- Jupyter Notebook `.ipynb` with all the steps clearly commented.
- The output of each step as specified above (e.g., head of DataFrame, info, describe, missing value counts, evaluation metrics for all models, coefficients, and intercept).
- Visualizations for outlier detection and residual analysis.
- A brief discussion answering the interpretation questions in sections 1d, 1e, 2e, 3c, 3d, 4b, and 4c.

Scratch paper

Name:

USC ID:

Scratch paper

Name:

USC ID:

Cumulative Distribution Function, $F(z)$, of the Standard Normal Distribution Table

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09

Cumulative Distribution Function, $F(z)$, of the Standard Normal Distribution Table

F - Distribution ($\alpha = 0.01$ in the Right Tail)

Denominator Degrees of Freedom	df ₂	df ₁	Numerator Degrees of Freedom								
			1	2	3	4	5	6	7	8	9
	1	4052.2	4999.5	5403.4	5624.6	5763.6	5859.0	5928.4	5981.1	6022.5	
	2	98.503	99.000	99.166	99.249	99.299	99.333	99.356	99.374	99.388	
	3	34.116	30.817	29.457	28.710	28.237	27.911	27.672	27.489	27.345	
	4	21.198	18.000	16.694	15.977	15.522	15.207	14.976	14.799	14.659	
	5	16.258	13.274	12.060	11.392	10.967	10.672	10.456	10.289	10.158	
	6	13.745	10.925	9.7795	9.1483	8.7459	8.4661	8.2600	8.1017	7.9761	
	7	12.246	9.5466	8.4513	7.8466	7.4604	7.1914	6.9928	6.8400	6.7188	
	8	11.259	8.6491	7.5910	7.0061	6.6318	6.3707	6.1776	6.0289	5.9106	
9	10.561	8.0215	6.9919	6.4221	6.0569	5.8018	5.6129	5.4671	5.3511		
10	10.044	7.5594	6.5523	5.9943	5.6363	5.3858	5.2001	5.0567	4.9424		
11	9.6460	7.2057	6.2167	5.6683	5.3160	5.0692	4.8861	4.7445	4.6315		
12	9.3302	6.9266	5.9525	5.4120	5.0643	4.8206	4.6395	4.4994	4.3875		
13	9.0738	6.7010	5.7394	5.2053	4.8616	4.6204	4.4410	4.3021	4.1911		
14	8.8616	6.5149	5.5639	5.0354	4.6950	4.4558	4.2779	4.1399	4.0297		
15	8.6831	6.3589	5.4170	4.8932	4.5556	4.3183	4.1415	4.0045	3.8948		
16	8.5310	6.2262	5.2922	4.7726	4.4374	4.2016	4.0259	3.8896	3.7804		
17	8.3997	6.1121	5.1850	4.6690	4.3359	4.1015	3.9267	3.7910	3.6822		
18	8.2854	6.0129	5.0919	4.5790	4.2479	4.0146	3.8406	3.7054	3.5971		
19	8.1849	5.9259	5.0103	4.5003	4.1708	3.9386	3.7653	3.6305	3.5225		
20	8.0960	5.8489	4.9382	4.4307	4.1027	3.8714	3.6987	3.5644	3.4567		
21	8.0166	5.7804	4.8740	4.3688	4.0421	3.8117	3.6396	3.5056	3.3981		
22	7.9454	5.7190	4.8166	4.3134	3.9880	3.7583	3.5867	3.4530	3.3458		
23	7.8811	5.6637	4.7649	4.2636	3.9392	3.7102	3.5390	3.4057	3.2986		
24	7.8229	5.6136	4.7181	4.2184	3.8951	3.6667	3.4959	3.3629	3.2560		
25	7.7698	5.5680	4.6755	4.1774	3.8550	3.6272	3.4568	3.3239	3.2172		
26	7.7213	5.5263	4.6366	4.1400	3.8183	3.5911	3.4210	3.2884	3.1818		
27	7.6767	5.4881	4.6009	4.1056	3.7848	3.5580	3.3882	3.2558	3.1494		
28	7.6356	5.4529	4.5681	4.0740	3.7539	3.5276	3.3581	3.2259	3.1195		
29	7.5977	5.4204	4.5378	4.0449	3.7254	3.4995	3.3303	3.1982	3.0920		
30	7.5625	5.3903	4.5097	4.0179	3.6990	3.4735	3.3045	3.1726	3.0665		
40	7.3141	5.1785	4.3126	3.8283	3.5138	3.2910	3.1238	2.9930	2.8876		
60	7.0771	4.9774	4.1259	3.6490	3.3389	3.1187	2.9530	2.8233	2.7185		
120	6.8509	4.7865	3.9491	3.4795	3.1735	2.9559	2.7918	2.6629	2.5586		
∞	6.6349	4.6052	3.7816	3.3192	3.0173	2.8020	2.6393	2.5113	2.4073		